

A Study in Transparent Adaptation and Joint-Level Feedback for Unsupervised Upper-Limb Rehabilitation

Mohit Madhu 

mohitmadhuj24@iiits.in

Human Computer Interaction Lab
Indian Institute of Information
Technology Sri City
Sri City, Andhra Pradesh, India

Chaitanya Anand 

chaitanyaanand.p24@iiits.in

Human Computer Interaction Lab
Indian Institute of Information
Technology Sri City
Sri City, Andhra Pradesh, India

Sambhav Manohar 

sambhav.m24@iiits.in

Human Computer Interaction Lab
Indian Institute of Information
Technology Sri City
Sri City, Andhra Pradesh, India

Abhijit Patil 

abhijitchandrakant.p23@gmail.com

Human Computer Interaction Lab
Indian Institute of Information
Technology Sri City
Sri City, Andhra Pradesh, India

Prajwal Gundala 

prajwal.g24@iiits.in

Human Computer Interaction Lab
Indian Institute of Information
Technology Sri City
Sri City, Andhra Pradesh, India

Himangshu Sarma 

himangshu.sarma@iiits.in

Human Computer Interaction Lab
Indian Institute of Information
Technology Sri City
Sri City, Andhra Pradesh, India

Gabriel Zachmann 

zach@cs.uni-bremen.de

University of Bremen
Mathematics and Computer Science
Bremen, Germany



Figure 1: XR Rehab, a mixed-reality application for upper-limb rehabilitation. (Left) The modular exercise selection interface, allowing clinicians to define and customize therapy routines such as tendon glides. (Right) Joint-level corrective feedback delivered during gamified therapy, shown here for a Flexion (Palm Down) exercise.

Abstract

This paper presents a design study exploring how extended reality (XR) systems can support unsupervised upper-limb rehabilitation through transparent adaptation and embodied feedback. This paper introduces XR Rehab, a framework instantiating two key design strategies: (1) explicit, clinician-configurable tolerance-based difficulty progression that makes adaptation logic visible and auditable, and (2) anatomically grounded joint-level corrective feedback that

embeds movement cues directly onto the user's tracked hand, reducing cognitive translation during active execution. Exercises are decomposed into modular gesture units with configurable feature weighting, enabling clinicians to tailor correctness criteria to individual patient needs. Through a user study with 21 healthy participants across 58 sessions, this study evaluates the design's impact on perceived workload and usability. Results indicate significant reductions in mental, physical, and temporal demand over repeated sessions without loss of perceived performance, while an A/B comparison demonstrates strong user preference for joint-level feedback over text-based cues. This paper discusses the implications of transparent adaptation and embodied feedback design for XR rehabilitation.

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1 Introduction

Extended reality (XR) offers promising opportunities for motor rehabilitation, yet its design for unsupervised home use remains underexplored. Two persistent design challenges limit deployment beyond clinical settings: (1) adaptive difficulty mechanisms that obscure progression logic from both clinicians and patients, and (2) feedback modalities that require users to cognitively translate abstract indicators into corrective movements during active exercise.

Building upon the modular architecture proposed in our preliminary work [14], this paper presents a design study addressing these challenges through the empirical evaluation of XR Rehab. We focus on two interconnected strategies: first, an explicit, clinician-configurable tolerance-based adaptation that makes difficulty progression mathematically visible and auditable; second, anatomically grounded joint-level feedback that visualizes corrective cues directly on the user's tracked hand skeleton, embedding guidance within embodied movement rather than abstract displays.

Importantly, this work is framed as an interaction design study rather than a clinical trial evaluating medical outcomes. Through an evaluation involving 21 healthy young adult participants across 58 gameplay sessions, this study assesses how these design choices influence perceived workload, usability, and user preference. By validating these interaction paradigms with non-impaired users, this study contributes foundational HCI insights for XR systems intended for future unsupervised clinical deployment.

2 Related Work

2.1 XR and Mixed Reality Systems for Upper-Limb Rehabilitation. XR has been widely studied for post-stroke upper-limb rehabilitation, with meta-analyses and systematic reviews reporting improved motor outcomes and patient engagement compared to conventional therapy [7, 11, 13, 16]. Adaptive MR systems demonstrated that combining physical reaching tasks with virtual feedback improves movement quality and functional outcomes [8, 9]. Recent MR platforms have sought to extend rehabilitation beyond the clinic into unsupervised home settings [17], broadening accessibility for patients who cannot attend frequent supervised sessions. However, most existing systems rely on predefined progression schemes with limited real-time adaptation to individual performance, and few provide clinicians with auditable control over how difficulty evolves across sessions.

2.2 Adaptive Difficulty and Personalization in Rehabilitation Systems. Adaptive rehabilitation adjusts task difficulty based on patient performance to maintain an appropriate level of therapeutic challenge. Early frameworks introduced performance-driven adaptation for telerehabilitation [2], establishing the importance of responsive difficulty control in remote settings. More recent

approaches have applied reinforcement learning [15] and Pareto-based dynamic difficulty adjustment [3] to balance competing rehabilitation objectives, often at the cost of transparency and clinician interpretability. A key limitation of many adaptive systems is that their progression logic remains opaque, making it difficult for clinicians to audit, configure, or trust the adaptation process. Accurate motion sensing is essential for reliable exercise evaluation underpinning any adaptive system; recent studies confirm that markerless hand tracking on the Meta Quest 2 provides sufficient kinematic accuracy for rehabilitation applications when properly calibrated [1, 6].

2.3 Gamification, Engagement, and Measurement Validity.

Gamification elements such as points, progression mechanics, and narrative contexts have been shown to improve motivation and long-term adherence in rehabilitation [4], making them a valuable complement to therapeutic exercise. However, integrating gameplay must be balanced carefully to avoid distorting prescribed movements or introducing unintended time pressure. Evaluating the impact of such systems requires validated measurement instruments. Perceived workload is commonly assessed using the NASA Task Load Index, though its theoretical and methodological application in HCI contexts warrants careful consideration [5]. System usability is frequently measured with the System Usability Scale, for which established benchmarks now exist specifically for digital health applications [12], enabling more meaningful interpretation of usability scores in rehabilitation technology evaluations.

3 Design Goals

Informed by interviews with three clinical experts (two physiotherapists, one rehabilitation specialist) who emphasized interpretable motion tracking, exercise diversity across anatomical regions, safe progression without time pressure, and auditable adaptation for remote monitoring, XR Rehab follows four design goals:

- **G1: Modular Decomposition** — Represent exercises as interpretable gesture units reusable across conditions.
- **G2: Clinician Control** — Enable transparent, clinician-configurable weighting instead of opaque learned models.
- **G3: Safe Unsupervised Progression** — Support gradual tolerance-based adaptation while preventing unsafe escalation.
- **G4: Engagement without Distortion** — Use gameplay to motivate participation without altering prescribed movements.

While instantiated for Carpal Tunnel Syndrome exercises¹, the framework is exercise-agnostic, generalizing to diverse anatomical regions without modification.

4 System Overview

XR Rehab maps tracked hand movements to therapeutic gesture representations and adaptive control parameters through a five-stage pipeline: hand tracking, gesture unit evaluation, weighted accuracy computation, adaptive tolerance updating, and gameplay control signal generation. Hand pose data is captured via markerless

¹Exercise protocols adapted from OrthoInfo (<https://orthoinfo.aaos.org/>), developed and reviewed by experts at the American Academy of Orthopaedic Surgeons.

tracking and converted into kinematic features (joint angles, inter-finger distances, orientation). Exercises are ordered sequences of gesture units specifying relevant joints and constraints, with accuracy computed as a weighted combination of features. Session-level performance updates tolerance parameters through an adaptive control loop. These tolerances define acceptable execution ranges and evolve across sessions, while sensing, evaluation, adaptation, and feedback remain decoupled.

5 Interaction Design

XR Rehab's interaction design is informed by the formative expert interviews, emphasizing motion accuracy, transparency, and interpretability. Clinicians positively assessed alignment between real-world movements and in-game responses, highlighting the importance of understanding how movement components contribute to correctness. Interactions are structured around decomposed gesture units, explicit kinematic features, and transparent weighting schemes rather than monolithic actions or opaque models. This enables safe unsupervised execution while allowing clinicians to influence correctness and progression without altering prescribed form.

5.1 Gesture Representation and Evaluation Design

Exercises are represented as sequences of gesture units rather than single actions. Each exercise is decomposed into clinically meaningful components, allowing focus on one unit at a time while preserving execution order. Each gesture unit defines relevant joints and movement features (rotations, distances, orientation). Progression occurs only when the current unit is completed within acceptable bounds, preventing compensation through unrelated movements. This keeps exercises modular while enabling fine-grained feedback and safe progression. Each gesture unit is evaluated using a weighted accuracy model, allowing joints and kinematic features to contribute unequally to correctness. All weights are clinician-configurable, enabling sensitive joints to be down-weighted based on pain, injury severity, or range-of-motion limits. Exposing weighting as a transparent control mechanism enables interpretable, patient-specific tuning and reuse of gesture structures across conditions, supporting safe personalized unsupervised rehabilitation.

5.2 Joint-Level Real-Time Corrective Feedback

Throughout the iterative design process, we observed that users often struggled to maintain precise recall of prescribed rehabilitation exercises once active gameplay began. Our initial approach relied on static reference visuals and text-based feedback, requiring users to shift attention away from embodied movement and cognitively map abstract indicators to their hand configuration. Reading textual labels and mapping them back to one's own hand under time pressure imposed a high attentional burden, limiting real-time correction effectiveness and increasing cognitive overhead.

To mitigate this, XR Rehab employs anatomically grounded joint-level corrective feedback overlaid directly on the tracked hand skeleton. Joints initialize as green, establishing a neutral baseline; irrelevant joints remain green, while relevant joints transition along

a red–green gradient based on their weighted accuracy contribution. This anatomically grounded feedback ensures that correctness cues are spatially localized, enabling immediate error perception and rapid self-correction without diverting attention from embodied action.

Crucially, green saturation is tied to the tolerance parameter τ , with full green achieved at the desired accuracy $a_d = 1 - \tau$ rather than perfect execution. Signaling success at a_d discourages over-exertion, as maximal accuracy or full-range execution may correspond to unsafe joint configurations. The green zone visually represents this tolerance, producing permissive, confidence-building feedback early in the process and progressively tightening as tolerance decreases.

To validate this approach, a within-subject A/B evaluation was conducted in which participants performed identical rehabilitation gestures using both text-based and joint-level feedback, with condition order counterbalanced. After completing both conditions, participants provided comparative preference judgments, Likert-scale usability ratings, and Raw NASA-TLX workload assessments for each feedback modality.

All participants ($N = 10$) preferred joint-level feedback (JLF) over text feedback for overall use, understanding gestures, lower mental effort, and gameplay integration (100% selection). Likert ratings favored JLF on key dimensions including gesture understanding (JLF = 5.8 vs Text = 3.6, $p = 0.0078$), mistake correction without guidance (5.7 vs 3.1, $p = 0.0020$), and reduced distraction (5.5 vs 3.8, $p = 0.0137$), with 95% CIs showing minimal overlap. RTLX analysis showed significantly lower overall workload for JLF ($M = 4.27$) compared to text feedback ($M = 5.73$; Wilcoxon $p = 0.0098$, rank-biserial $r = 0.95$). Subscale analysis indicated reduced mental demand ($p = 0.031$), effort ($p = 0.031$), and frustration ($p = 0.037$), with no significant differences in physical or temporal demand. These results indicate that anatomically grounded joint-level cues substantially reduce cognitive load while improving real-time corrective effectiveness. Figure 2 illustrates the joint-level feedback visualization and summarizes the corresponding A/B workload comparison.

5.3 Onboarding, Failure, and Recovery Design

Failure is designed as recoverable rather than punitive. When accuracy falls below the threshold, health-point loss is introduced gradually, allowing continued adjustment. If health points reach zero, nearby enemies are cleared, and the current gesture unit restarts. This prevents progression with unsafe movements while avoiding discouraging penalties. Restarting only the active unit maintains therapeutic pacing and focuses correction on specific errors. The tutorial supports user familiarization and system calibration during onboarding. Users are guided through gestures with demonstrations and cues, encouraged to explore without performance pressure. Joint-level feedback uses a neutral blue gradient instead of red–green to avoid premature success notions. Kinematic data collected during onboarding establishes gesture-specific baselines, initializing tolerance parameters relative to demonstrated capability rather than fixed standards.

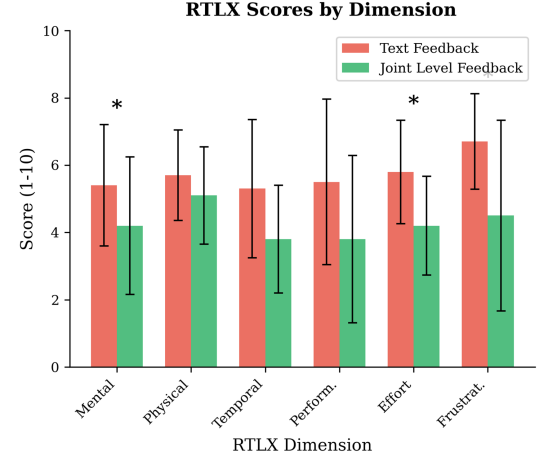
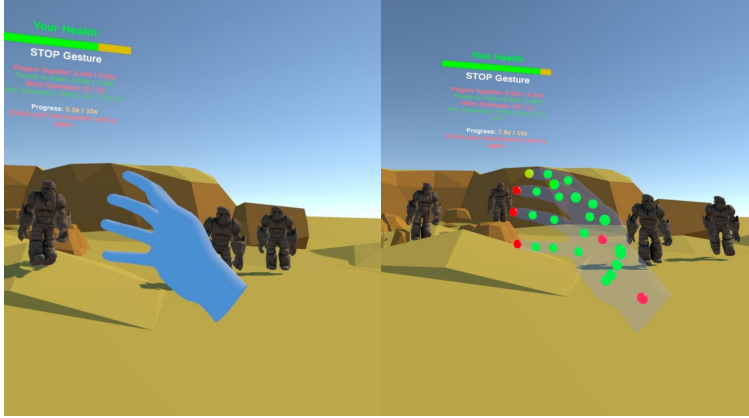


Figure 2: A/B comparison of text-based and joint-level corrective feedback. (a) Example in-game feedback visualizations, illustrating static text cues versus anatomically grounded joint-level feedback overlaid on the tracked hand skeleton. (b) Raw NASA-TLX workload scores by dimension (mean ± 95% CI). Joint-level feedback significantly reduces perceived mental demand and effort compared to text-based feedback (paired Wilcoxon tests, $p < 0.05$), with no significant differences observed in physical demand, temporal demand, perceived performance, or frustration.

6 Tolerance Algorithm

Building upon the modular adaptive framework introduced by Manohar et al. [14], XR Rehab employs an explicit tolerance-based adaptation rule. While prior work proposed a preliminary linear heuristic for thresholding, this study introduces a refined, bounded performance signal P that normalizes performance relative to current expectations, ensuring stable, long-term adaptation.

For each gesture unit, the system maintains a tolerance parameter $\tau \in (0, 1)$ defining the acceptable deviation from ideal execution, with a desired accuracy of $a_d = 1 - \tau$. After each session, the aggregated execution accuracy $a \in [0, 1]$ is computed. The normalized performance signal $P \in (-1, 1)$ is defined as:

$$P = \frac{a - a_d}{\tau + |a - a_d|} \quad (1)$$

The sign of P indicates overperformance ($P > 0$) or underperformance ($P < 0$). Normalization by τ ensures that identical absolute errors produce smaller adaptation signals as tolerance tightens, reacting to consistency relative to current expectations rather than raw precision. Tolerance is then updated multiplicatively:

$$\tau_{next} = \tau(1 - (1 - c)P) \quad (2)$$

Tolerance updates occur at the session level to aggregate performance, naturally attenuating transient tracking noise and momentary lapses.

6.1 Adaptability Coefficient

The adaptability coefficient $c \in (0, 1]$ explicitly modulates progression speed based on clinical context. It is defined as a clinician-configured weighted combination of demographic (C_{demo}) and injury severity ($C_{severity}$) factors:

$$c = w_d C_{demo} + w_s C_{severity} \quad (3)$$

(where $w_d + w_s = 1$). This explicit formulation addresses the need for clinical interpretability. It allows therapists to safely slow progression (higher c) for higher-risk profiles or accelerate it (lower c) for lower-risk users. Overperformance smoothly tightens tolerance (increasing challenge), while underperformance relaxes it to prevent unsafe escalation.

Practical Examples of Adaptation: To illustrate the bidirectional tolerance updates and personalized regulation of the of the algorithm through the use of tolerance and adaptability coefficient respectively, consider a baseline where a patient's target accuracy is $a_d = 0.80$ (thus $\tau = 0.20$):

- **Scenario A (Overperformance, Moderate Risk):** With an adaptability coefficient of $c = 0.5$, the patient achieves an actual accuracy of $a = 0.90$. The performance signal evaluates to $P = 0.33$. The updated tolerance tightens to $\tau_{next} \approx 0.167$. The next session's target accuracy smoothly increases to 83.3%.
- **Scenario B (Underperformance, Moderate Risk):** Given the same $c = 0.5$, if the patient struggles and achieves $a = 0.70$, the signal becomes $P = -0.33$. The tolerance safely relaxes to $\tau_{next} \approx 0.233$, lowering the target accuracy to 76.7% to prevent unsafe escalation or frustration.
- **Scenario C (Overperformance, High Clinical Risk):** If the patient has a severe injury profile yielding a conservative $c = 0.9$, the same overperformance of $a = 0.90$ ($P = 0.33$) results in a heavily dampened update. The tolerance only tightens to $\tau_{next} \approx 0.193$. The new target accuracy becomes 80.7%, demonstrating how the algorithm enforces cautious progression for vulnerable patients despite immediate in-game success.

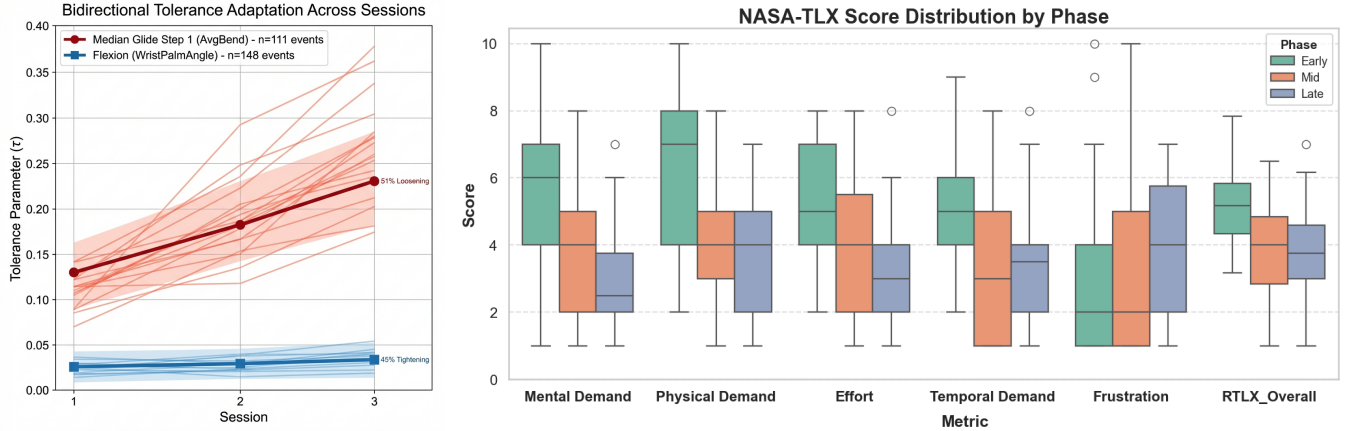


Figure 3: Perceived workload and adaptive tolerance across study phases. (A) Tolerance trajectories show bidirectional adaptation, with loosening and tightening patterns at individual (thin lines) and group (bold lines) levels. (B) Distribution of workload scores (NASA-RTLX) across Early, Mid, and Late phases. Overall workload decreases from Early to Late, with similar trends in Mental, Physical, and Temporal Demand, while Frustration and inverted Performance remain stable.

6.2 Relation to the Power Law of Practice

Furthermore, across repeated sessions, this multiplicative tolerance trajectory exhibits rapid early change followed by gradual convergence. Because updates scale with the current τ , larger adjustments occur when tolerance is initially loose. As the user improves and τ tightens, both the tolerance magnitude and performance variance decrease, yielding smaller, stable updates. This emergent behavior naturally aligns with the well-documented power law of practice in motor learning, demonstrating that the adaptation mechanism structurally supports human motor skill acquisition while remaining entirely clinician-auditable.

7 Evaluation

8 Implementation & Gameplay Design

XR Rehab embeds gesture evaluation and tolerance-based adaptation within wave-based gameplay designed to provide mandatory rest between exercises. Gameplay progresses through discrete waves, each requiring completion of prescribed gesture units, separated by recovery periods that support controlled pacing and fatigue management. Progression is driven by gesture execution rather than speed or reflexes; incorrect execution triggers recovery, preventing unsafe advancement. Game elements maintain engagement but do not modify gesture requirements or bypass therapeutic constraints. Session-level metrics are logged for clinician review and longitudinal monitoring.

8.1 Participants and System Usage

Twenty-one healthy participants (16M/5F, age $M = 19.9$, $SD = 7.6$) completed 58 sessions over one week. Participation was not enforced to be identical; 16 participants completed 3 sessions while 5 completed 2 sessions, reflecting ecological variability in adherence. This variance allows us to robustly assess tolerance adaptation

stability across different exposure durations. Mean session accuracy was stable at 83.0% ($SD = 2.8\%$), with per-gesture difficulty ranging from 45.2% (Wrist Extension) to 96.7% (Tendon Glides), validating the need for granular adaptation. Analysis of 1,073 tolerance events showed 44.5% tightening and 24.0% loosening, with individual patterns varying from 12% to 79% tightening across participants. High-variance gestures (Flexion: $SD = 14.9\%$; Median Glide Step 2: $SD = 14.3\%$) indicate substantial individual differences in motor control, confirming the necessity of personalized tolerance parameters.

8.2 Workload Evaluation (RTLX)

Perceived workload was assessed using the NASA Raw Task Load Index (RTLX) [10], aggregated into Early, Mid, and Late phases based on chronological session order. Descriptively, Mental Demand, Physical Demand, Temporal Demand, Effort, and overall RTLX all decreased from Early to Late phases, while inverted Performance and Frustration remained stable (Fig. 3). For example, Mental Demand decreased from $M = 5.69$ ($SD = 2.29$) in Early sessions to $M = 3.00$ ($SD = 1.75$) in Late sessions, and overall RTLX decreased from $M = 5.11$ ($SD = 1.17$) to $M = 3.85$ ($SD = 1.60$). Wilcoxon signed-rank tests with Holm–Bonferroni correction showed significant Early–Late reductions for Mental Demand, Physical Demand, Temporal Demand, Effort, and overall RTLX ($p_{\text{corr}} < .05$ for all), with no significant changes in Frustration or inverted Performance. Friedman tests on participants with complete phase data ($n = 14$) confirmed significant phase effects for the same dimensions ($p < .01$), with medium-to-large effect sizes (Kendall’s $W = 0.37$ – 0.59). These results indicate reduced cognitive and physical workload over repeated use without loss of perceived performance, suggesting improved interaction efficiency rather than disengagement.

8.3 System Usability Scale (SUS) Evaluation

SUS was administered after initial exposure to assess first-use usability and learnability. Across $N = 33$ participants, the system achieved a mean SUS score of 65.00 ($SD = 16.99$; 95% CI [59.20, 70.80]), indicating acceptable but below-average usability.

However, it is critical to contextualize this metric within the deployment constraints of unsupervised rehabilitation. Because the SUS was collected prior to extended practice, these results primarily reflect first-use usability, initial confidence, and learnability rather than optimized expert performance. The relatively large score variance suggests heterogeneous first-time experiences, which is expected given differences in participants' prior XR familiarity.

Given that the system targets unsupervised home use, where clinical supervision is absent, first-use learnability is paramount. This baseline score suggests that while the interaction paradigm is functionally viable, the initial cognitive overhead of learning gesture-based spatial interactions remains high for first-time users. Consequently, future iterations must prioritize interface refinement, visual fidelity, and robust onboarding tutorials to better accommodate users with lower XR exposure.

9 Conclusion

This design study investigated transparent adaptation and embodied feedback for unsupervised XR rehabilitation exergames. Guided by four clinical design goals, this study developed explicit tolerance-based difficulty progression and anatomically grounded joint-level corrective feedback within a modular gesture unit framework. Our evaluation with 21 participants across 58 sessions demonstrated stable performance, significant workload reductions (28–42% across RTLX subscales), and unanimous preference for joint-level over text-based feedback. Key limitations include healthy young adult participants, limited longitudinal window (2–3 sessions), and controlled deployment settings. Future work should pursue clinical validation with patient populations, multi-week unsupervised home studies, clinician-facing monitoring dashboards, and extension to additional anatomical regions. This paper argues that transparent adaptation and embodied feedback represent promising design directions for XR rehabilitation systems.

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References

- [1] Diar Abdulkarim et al. 2024. A methodological framework to assess the accuracy of virtual reality hand-tracking systems: a case study with the Meta Quest 2. *Behavior Research Methods* 56, 2 (2024), 1052–1063. doi:10.3758/s13428-022-02051-8
- [2] Imad Afyouni, Anas Einea, and Abdullah Murad. 2019. Towards an Adaptive Gaming Framework for TeleRehabilitation. In *Proceedings of the 12th ACM International Conference on Pervasive Technologies Related to Assistive Environments (PETRA '19)*. Association for Computing Machinery, New York, NY, USA. doi:10.1145/3316782.3321545
- [3] Oladayo S. Ajani and Rammohan Mallipeddi. 2023. Pareto-based Dynamic Difficulty Adjustment of a competitive exergame for arm rehabilitation. *International Journal of Human-Computer Studies* 178 (2023), 103100. doi:10.1016/j.ijhcs.2023.103100
- [4] Fabio Alfieri et al. 2022. Gamification in musculoskeletal rehabilitation: a narrative review. *Current Reviews in Musculoskeletal Medicine* 15, 6 (2022), 629–636. doi:10.1007/s12178-022-09797-w
- [5] Ebrahim Babaei, Tilman Dingler, Benjamin Tag, and Eduardo Velloso. 2025. Should we use the NASA-TLX in HCI? A review of theoretical and methodological issues around mental workload measurement. *International Journal of Human-Computer Studies* 201 (2025), 103515. doi:10.1016/j.ijhcs.2025.103515
- [6] D. Borzelli, V. Boarini, and A. Casile. 2025. A quantitative assessment of the hand kinematic features estimated by the Oculus Quest 2. *Scientific Reports* 15 (2025), 8842. doi:10.1038/s41598-025-91552-5
- [7] Matteo Ceradini, Elena Losanno, Silvestro Micera, Andrea Bandini, and Silvia Orlandi. 2024. Immersive VR for upper-extremity rehabilitation in patients with neurological disorders: a scoping review. *Journal of NeuroEngineering and Rehabilitation* 21 (2024), 75. doi:10.1186/s12984-024-01367-0
- [8] Margaret Duff, Yinpeng Chen, Suneth Attygalle, Janice Herman, Hari Sundaram, Gang Qian, Jiping He, and Thanassis Rikakis. 2010. An adaptive mixed reality training system for stroke rehabilitation. *IEEE Transactions on Neural Systems and Rehabilitation Engineering* 18, 5 (2010), 531–541. doi:10.1109/TNSRE.2010.2055061
- [9] Margaret Duff, Yinpeng Chen, Long Cheng, Sheng-Min Liu, Paul Blake, Steven L. Wolf, and Thanassis Rikakis. 2013. Adaptive mixed reality rehabilitation improves quality of reaching movements more than traditional reaching therapy following stroke. *Neurorehabilitation and Neural Repair* 27, 4 (2013), 306–315. doi:10.1177/1545968312465195
- [10] Sandra G. Hart and Lowell E. Staveland. 1988. Development of NASA-TLX (Task Load Index): Results of Empirical and Theoretical Research. In *Human Mental Workload*. 139–183. doi:10.1016/S0166-4115(08)62386-9
- [11] Jiaqi Huang et al. 2025. Effect of Home-Based Virtual Reality Training on Upper Extremity Recovery in Patients With Stroke: Systematic Review. *Journal of Medical Internet Research* 27 (2025), e69003. doi:10.2196/69003
- [12] Maciej Hyzy, Raymond Bond, Maurice Mulvenna, Lu Bai, Alan Dix, Simon Leigh, and Sophie Hunt. 2022. System Usability Scale Benchmarking for Digital Health Apps: Meta-analysis. *JMIR mHealth and uHealth* 10, 8 (2022), e37290. doi:10.2196/37290
- [13] Kate Laver et al. 2025. Virtual reality for stroke rehabilitation. *Cochrane Database of Systematic Reviews* 2025, 6 (2025), CD008349. doi:10.1002/14651858.CD008349.pub5
- [14] Sambhav Manohar, Mohit Madhu, Chaitanya Anand, Abhijit Patil, Gabriel Zachmann, and Himangshu Sarma. 2025. Towards a Modular XR Platform for Personalized Upper Limb Rehab. In *2025 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*. IEEE, 887–888. doi:10.1109/ISMAR-Adjunct68609.2025.00240
- [15] Avishag Deborah Pelosi et al. 2024. Personalized rehabilitation approach for reaching movement using reinforcement learning. *Scientific Reports* 14 (2024), 17675. doi:10.1038/s41598-024-64514-6
- [16] Margarida F. Pereira et al. 2020. Application of AR and VR in hand rehabilitation: A systematic review. *Journal of Biomedical Informatics* 111 (2020), 103584. doi:10.1016/j.jbi.2020.103584
- [17] Qi Wang, Can Luo, Shuai Yang, Shuo Xu, Yibo Yang, Jie Jia, and Bin Yu. 2024. MRRehab: A Mixed Reality Rehabilitation System Supporting Integrated Speech and Hand Training. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 8, 3, Article 133 (2024), 23 pages. doi:10.1145/3678532